LightRAG: A Lightweight Retrieval-Augmented Framework for Page-Aware Academic Question Answering

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Abstract

Academic literature question answering requires answers that are both accurate and verifiable. We present LightRAG, a lightweight retrieval-augmented framework that combines page-aware chunking, hybrid dense-sparse retrieval, cross-encoder reranking, and citation-grounded generation. LightRAG splits documents along page and section boundaries, retrieves candidates with Qwen3-Embedding and BM25, fuses their scores with a weighted sum, and reranks the top candidates with a cross-encoder before generation. On two synthetic benchmarks, SciQA-Demo and PaperQA-Demo, LightRAG achieves an F1 of 0.82, outperforming BM25-RAG (0.69) and a dense-only baseline (0.74). An ablation study shows that page-aware chunking and reranking contribute 6.1 and 4.7 F1 points, respectively. The system answers in 1.8 seconds on average on a single consumer GPU. These results suggest that lightweight hybrid retrieval is a practical foundation for trustworthy literature assistants.

Keywords

retrieval-augmented generation; academic question answering; hybrid retrieval; cross-encoder reranking; citation grounding

1. Introduction

Researchers increasingly rely on large language models to skim, summarize, and compare academic papers. A general-purpose model, however, may produce fluent statements that are not supported by the source document. In a literature review, an unsupported sentence is worse than a missing one: it can send a reader to the wrong method, dataset, or conclusion.

Retrieval-augmented generation (RAG) addresses this problem by retrieving passages before generation. Yet three practical issues remain. First, fixed-size chunking destroys page and section boundaries, so a citation cannot reliably point back to the original page. Second, dense retrieval captures paraphrase but often misses exact terms such as model names, dataset names, and symbols. Third, a single retrieval stage returns noisy candidates, and the generator may over-trust them.

This demo report introduces LightRAG, a deliberately small framework that targets these issues. The main contributions are:

• Page-aware chunking that preserves page, section, and block-type metadata for every chunk.

• Hybrid retrieval that fuses dense and sparse scores, followed by cross-encoder reranking.

• A citation-grounded prompt that requires every factual claim to carry a page reference.

• A synthetic evaluation that reports F1, citation accuracy, latency, and an ablation study.

The remainder of this report describes related work, the method, the experimental setup, results, limitations, and conclusions. All data in this document are fictional and were created solely to demonstrate a document-reading interface.

2. Related Work

2.1 Retrieval-Augmented Generation

RAG combines a retriever with a generator. The retriever selects passages from a corpus, and the generator conditions its answer on those passages. This design reduces the need to store factual knowledge in model parameters and makes it possible to attach evidence to an answer. However, the quality of the final answer is bounded by the quality of retrieval.

2.2 Sparse and Dense Retrieval

BM25 is a strong sparse baseline. It rewards exact term overlap and remains competitive when queries contain rare names or symbols. Dense retrievers such as DPR and Contriever map queries and passages into a shared vector space and handle paraphrase better. The two families fail in different ways, which motivates score fusion rather than a single retriever.

2.3 Cross-Encoder Reranking

A cross-encoder reads a query and a candidate passage jointly and produces a relevance score. It is more accurate than a bi-encoder but too expensive to run over an entire corpus. A common compromise is to retrieve a moderate candidate set and rerank only that set. In this demo, fifty candidates are reranked down to six passages.

2.4 Citation-Grounded Question Answering

Prior work has explored sentence-level attribution and post-hoc citation verification. These methods are valuable but add another model to the pipeline. LightRAG instead keeps page identifiers in the retrieved context and instructs the generator to cite them directly. This is simpler to inspect and easier to evaluate in a document reader.

3. Method

3.1 Page-Aware Chunking

The parser walks through the PDF page by page. Text is split at section headings and then packed into chunks of at most 512 tokens with a 20 percent overlap. Every chunk stores its page number, section title, and block type. Tables and displayed formulas are kept as separate chunks so that their structure is not merged into surrounding prose. This metadata is later copied into the citation attached to each answer.

3.2 Hybrid Retrieval

Given a question q and a candidate chunk d, LightRAG computes a dense score with Qwen3-Embedding and a sparse score with BM25. Both scores are normalized to the interval [0, 1] and combined with a weighted sum:

$$S(q, d) = \alpha \cdot S_{dense}(q, d) + (1 - \alpha) \cdot S_{bm25}(q, d)$$

Unless stated otherwise, alpha is set to 0.6, which gives the dense retriever slightly more weight while retaining exact-term matching. BM25 uses k1 = 1.5 and b = 0.75. The fusion produces fifty candidates per question. A sensitivity check in Section 5 shows that alpha = 0.5 and alpha = 0.7 perform within 0.8 F1 points of the default.

3.3 Cross-Encoder Reranking

The fifty fused candidates are passed to a bge-reranker-base cross-encoder. The reranker scores each query-passage pair jointly and keeps the top six passages. On the demo GPU this step takes about 12 milliseconds per query. Reranking is especially helpful for questions that mention a model name or a numeric result, because the cross-encoder can compare the exact wording of the question with the candidate passage.

3.4 Citation-Grounded Generation

The generator receives the question, the six reranked passages, and a strict instruction: every factual sentence must end with a page reference in the form [p.N]. If the passages do not contain enough evidence, the model must say so instead of guessing. The temperature is set to 0.2. A post-processing check parses the cited page numbers and highlights them in the reader; citations that do not correspond to a supplied passage are discarded.

Algorithm 1: LightRAG query pipeline



Two design choices are worth emphasizing. First, explicit page questions bypass semantic retrieval entirely: if a user asks about page 7, the system reads page 7. Second, the generator is not allowed to invent a page number. This constraint makes the interface predictable and gives the reader a direct way to verify each statement.

4. Experiments

4.1 Datasets and Metrics

SciQA-Demo contains 1,200 question-answer pairs built from 40 synthetic papers. PaperQA-Demo contains 800 pairs from 25 synthetic papers. Every question is annotated with the page that contains the answer. We report exact match (EM), token-level F1, citation accuracy (CA), and average latency. CA counts a citation as correct when the cited page overlaps the annotated evidence page.

4.2 Baselines

BM25-RAG uses sparse retrieval only. DPR uses a dense retriever without reranking. RAG-base uses dense retrieval with the same generator but no page-aware metadata. LightRAG uses hybrid retrieval and reranking. All methods use the same six-passage context budget.

Table 1: Main results on the two synthetic benchmarks

Method	Retrieval	F1	CA	Latency
BM25-RAG	Sparse	0.69	0.61	0.9 s
DPR	Dense	0.74	0.66	1.3 s
RAG-base	Dense	0.78	0.71	1.5 s
LightRAG	Hybrid + rerank	0.82	0.79	1.8 s

F1 is computed from precision P and recall R:

$$F1 = 2 * P * R / (P + R), \quad P = TP / (TP + FP), \quad R = TP / (TP + FN)$$

LightRAG improves F1 by 0.04 over the strongest dense baseline and by 0.13 over BM25-RAG. The gain in citation accuracy is larger, which is important for a reading interface: a reader can verify an answer only if the cited page actually contains the evidence.

5. Results and Discussion

5.1 Ablation Study

Table 2 removes one component at a time. Page-aware chunking has the largest effect on citation accuracy, while reranking has the largest effect on F1 for questions containing exact model or dataset names.

Table 2: Ablation results (F1)

Variant	F1	CA
LightRAG (full)	0.82	0.79
w/o reranking	0.773	0.72
w/o page-aware chunking	0.759	0.64
w/o hybrid (dense only)	0.781	0.70

Removing reranking costs 4.7 F1 points. Removing page-aware chunking costs 6.1 F1 points and 15 citation-accuracy points, confirming that page metadata is not cosmetic. Replacing hybrid retrieval with dense-only retrieval costs 3.9 F1 points, mostly on questions that mention a precise model name or a numeric result.

5.2 Error Analysis

We manually inspected 100 incorrect answers. Retrieval missed the evidence in 38 cases, OCR noise affected 27 cases, multi-hop questions accounted for 21 cases, and generation errors accounted for 14 cases. The first two categories suggest that better page parsing and OCR confidence filtering are more valuable than a larger generator.

5.3 Discussion

The latency cost of LightRAG is 0.4 seconds relative to RAG-base. In a reading interface this is acceptable because the answer streams token by token and the user can continue reading while the response is produced. The results support a simple claim: keeping page structure and combining two retrieval signals is a practical way to make a literature assistant more trustworthy without a large increase in system complexity.

6. Limitations and Conclusion

Limitations

The benchmarks are English-only and synthetic.

Results may not transfer to other languages or domains.

Scanned pages depend on OCR and may contain recognition errors.

Low-confidence OCR text is not filtered in this demo.

Long documents use only the six highest-ranked chunks.

Multi-hop questions across distant pages remain difficult.

The system needs a local embedding model and a hosted LLM.

Conclusion

LightRAG improves answer quality and citation accuracy with a small latency cost.

Future Work

Future work includes multilingual benchmarks, OCR confidence filtering, and user studies.